Chapter 1: Introduction to DBMS

1. What is Data?

Data is a collection of raw facts and figures that by themselves have little meaning.

Examples

- Name: Vrushali
- Age: 23
- Marks: 89
- Salary: ₹50,000

These are simply values until they are organized and processed.

Data + Processing = Information

2. What is Information?

Information is processed, organized, and meaningful data that helps in decision-making.

Example:

Data	Information
45, 67, 89	Average marks = 67
Sales values	Monthly Sales Report

3. What is a Database?

A **Database** is an organized collection of related data stored electronically so it can be accessed, managed, updated, and retrieved efficiently.

Definition

A database is a structured collection of logically related data.

Real-Life Examples

- College Management System
- Banking System
- Railway Reservation
- Instagram
- WhatsApp
- Amazon
- Hospital Management
- Library Management

4. What is DBMS?

DBMS = Database Management System

A DBMS is software that allows users to create, store, retrieve, update, and manage databases.

Definition

> DBMS is software that acts as an interface between the database and the user/application.

Instead of directly accessing files, users interact with the DBMS.

User
↓
Application
↓
DBMS
↓
Database

Examples:

- MySQL
- Oracle
- PostgreSQL
- Microsoft SQL Server
- SQLite
- MariaDB

5. Database vs DBMS

Database	DBMS
Collection of data	Software that manages data
Stores information	Creates, updates, deletes, retrieves data
Passive	Active software
Example: Student records	Example: MySQL

6. File System vs DBMS

File System	DBMS
Data stored in files	Data stored in databases
High redundancy	Low redundancy
Poor security	Strong security
Difficult searching	Easy searching using SQL
No relationships	Supports relationships
Limited backup	Backup & recovery
Data inconsistency	Consistent data
Less scalable	Highly scalable

14. Structured vs Semi-Structured vs Unstructured

Feature	Structured	Semi-Structured	Unstructured
Format	Fixed	Flexible	None
Schema	Yes	Partial	No
SQL Support	Yes	Limited	No
Examples	Tables	JSON, XML	Images, Videos
Easy Searching	Yes	Moderate	Difficult

Placement Interview Questions

Q1. What is a database?

A database is an organized collection of logically related data that can be easily accessed, managed, and updated.

Q2. What is DBMS?

DBMS is software used to create, manage, retrieve, update, and control databases.

Q3. Why is DBMS preferred over a file system?

Because it reduces redundancy, improves security, maintains consistency, supports backup and recovery, and allows concurrent access.

Q4. What is the difference between data and information?

- Data: Raw facts and figures.
- Information: Processed and meaningful data.

Q5. Name different types of databases.

- Hierarchical
- Network

- Relational
- Object-Oriented
- NoSQL

Q6. What is structured data?

Structured data is organized into rows and columns with a predefined schema.

Q7. Give examples of unstructured data.

Images, videos, audio files, emails, PDFs, and social media posts.

Q8. What is semi-structured data?

Data that doesn't follow a strict relational schema but contains organizational tags or key-value pairs, such as JSON and XML.